

# NLP Project 4

## BERT

(Bi-directional Encoder Representation from Transformers)

Detailed steps are given in last page.

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# Overview

- There will be 4 projects in total:
  - Project 1 and 2: Word2Vec
    - Project 1: CBOW; Project 2: Skip-Gram
  - Project 3: Transformer
  - Project 4: BERT

# BERT

- BERT (Bidirectional Encoder Representations from Transformers) is a groundbreaking pre-trained language model developed by Google in 2018. Based on the Transformer architecture, it captures deep bidirectional context by training on large text corpora using masked language modeling (MLM) and next sentence prediction (NSP). Unlike previous models, BERT dynamically generates contextualized word embeddings, enabling superior performance in NLP tasks like text classification, question answering, and named entity recognition. Its variants (e.g., BERT-Large, DistilBERT) balance power and efficiency, making it a cornerstone of modern NLP.

**We down load a small pretrained Bert model with 4 encoder layers.**

**We then fine tune the model, i.e., train the model starting Bert parameters on new data sets. So the model parameters changed slightly, more aligned with the new data set.**

# Dataset

- **The Stanford Sentiment Treebank (SST)** is a benchmark dataset for fine-grained sentiment analysis, providing sentiment labels at both sentence and phrase levels.

|                                                                                                                       |            |
|-----------------------------------------------------------------------------------------------------------------------|------------|
| it 's a charming and often affecting journey .                                                                        | 1 positive |
| unflinchingly bleak and desperate                                                                                     | 0 negative |
| allows us to hope that nolan is poised to embark a major career as a commercial yet inventive filmmaker .             | 1 positive |
| the acting , costumes , music , cinematography and sound are all astounding given the production 's austere locales . | 1 positive |
| it 's slow -- very , very slow .                                                                                      | 0 negative |

# Dataset

- **Microsoft Research Paraphrase Corpus** consists of sentence pairs automatically extracted from online news sources, with human annotations for whether the sentences in the pair are semantically equivalent, 1 means equivalent while 0 means not. Some examples are as follows:

|                                                                                 |                                                                                                                      |                  |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------|
| The AFL-CIO is waiting until October to decide if it will endorse a candidate . | The AFL-CIO announced Wednesday that it will decide in October whether to endorse a candidate before the primaries . | 1 equivalent     |
| No dates have been set for the civil or the criminal trial .                    | No dates have been set for the criminal or civil cases , but Shanley has pleaded not guilty .                        | 0 not_equivalent |

# Training Task

- The two Python files are named after the tasks and serve as the runner of the Bert model on their corresponding tasks. They will automatically download the task datasets and a small pretrained Bert model with 4 encoder layers from the internet, finetune the model for 2 epochs on the training set, and evaluate its performance (accuracy for mrpc, accuracy, and F1 for sst2) on the evaluation set at the end of each epoch. The progress bar on the command line shows the progress of the whole training process, not a single epoch. The model will be saved to the local path after training. Finally, 5 practical examples are provided where the trained model is used to solve the task.

# Evaluation&Output

```
{'eval_loss': 0.3737373650074005, 'eval_accuracy': 0.8440366972477065, 'eval_runtime': 0.2407, 'eval_samples_per_second': 3622.989, 'eval_steps_per_second': 116.335, 'epoch': 2.0}  
{'train_runtime': 120.8693, 'train_samples_per_second': 1114.411, 'train_steps_per_second': 34.831, 'train_loss': 0.33094014238008695, 'epoch': 2.0}
```

Practical task examples:

hide new secretions from the parental units

Prediction: 0, which is correct.

contains no wit , only labored gags

Prediction: 0, which is correct.

that loves its characters and communicates something rather beautiful about human nature

Prediction: 1, which is correct.

remains utterly satisfied to remain the same throughout

Prediction: 1, which is wrong.

on the worst revenge-of-the-nerds clichés the filmmakers could dredge up

Prediction: 0, which is correct.





# Project details/steps.

- Configure the relevant environment according to the requirements in the document and complete the corresponding training tasks
- Include the performance numbers after the final epoch on the evaluation set and the screenshot of the practical examples in your report.
- Note that the training set of mrpc is rather small compared to that of sst2, so it should be the first choice for you to run to examine the program's functionality.
- Running the model successfully : 40% of grades,
- Evaluation of two nlp tasks:
  - In sentiment analysis, pick up 4 sentences that two are classified correctly and another 2 classified incorrectly. Make some comments on that. 30% grade.
  - On the paraphrase task, pick 4 paragraph pairs, 2 of them are correct paraphrase and other 2 are incorrect. Make some discussions. 30% of grade.